

America passed the Digital Millenium Copyright Act (DMCA), an act that apparently criminalises the circumvention of DRM protection, even when there is no infringement of copyright itself. In 2001, the European Parliament passed the European Union Copyright Directive, which addresses some of the concerns of the US DMCA, but leaves one of its principle issues, that of limiting the liability of ISPs to prosecution, to the separate Electronic Commerce Directive. The original form of the DMCA, which illustrates just how convoluted such things can become, appeared to make cryptographic research illegal.

In 2006, the Indian Copyright office posted proposals to amend the Copyright Act of 1957 (last amended in 1999) on its website. One of the proposed amendments seeks to introduce the issue of Digital Rights Management (DRM) into Indian copyright law along the lines of the DMCA.

All of these methods of protection are primarily passive and defensive. Digital technology has finally made it relatively easy for copyright holders to protect their property by active and offensive means. At their most extreme, these measures make it possible to automatically punish anyone who exceeds the limits of the copyright and may no longer adhere to legal, but to almost arbitrary limits, chosen at the whim of the copyright holder and further, to analyse, manipulate and control its customer base.

A further difficulty for the international media companies is that copyright law varies from country to country. In some countries it is perfectly legal to make multiple copies ‘for personal use’. The term ‘fair use’, it seems, is often mis-quoted; in the US it refers to the right to quote short exerts of a copyrighted work for the purposes of commentary, criticism, news reporting, research and other similar purposes. The Digital Millennium Copyright Act (DMCA) has actually reduced the scope of ‘fair use’.

Prosecution of the owners of web sites supporting peer-to-peer networks offering access to illegal music downloads is hampered by the imprecision of current copyright law. Peer-to-peer technology itself is not illegal and the argument centres around whether simply making copyright material available, in a way that makes it easy to copy, can be deemed an illegal act. This has led in some cases to big media making highly publicised prosecutions of individuals they suspect of having made large numbers of illegal downloads.

The arguments over copyright oscillate between two extremes: making any copies of any kind of a copyright work is an infringement of copyright, or – copying of a copyright work by an individual who has paid for a copy, into different formats in order that that indi-

vidual may make use of the work, is reasonable and permissible. The problem with the first position is that digital technologies routinely make copies as part of their function, for example when reading a web page, the page is copied or cached.

Piracy and the big media companies

The big media companies seem to refuse to believe that falling sales are anything to do with natural market cycles, or changes in society, or to accept that just possibly their own reactionary business practices – diminishing interest in developing new artists, reliance on a few guaranteed high return headliners and re-selling old back catalogue as re-mastered on new media – have caused loss of interest. Instead, they say, it has to be piracy that’s bleeding the life blood out of the already moribund industry. Around eight years ago music industry analysts noticed increasingly sharp drop off’s in sales, following the release of a new CD. In the nineties, second week sales would typically fall by less than 25 per cent. In 2002, second week sales were dropping by over 40 per cent. The music industry took this as a clear sign that small scale piracy – people making copy CDs for their friends – was responsible. This is despite the fact that the same argument – that sales of vinyl LPs were suffering because people were copying them to tape – had been used to raise an indiscriminate blanket tax on blank cassette tapes, back in the 1970s at a time when the music industry was experiencing rapid growth. In India it seems the cassette tape actually created a booming market for pop music.

As we reported in a previous feature (February 2007 issue) on peer-to-peer technology, it has been suggested that DRM should be abandoned and a similar tax or rather levy (to be accurate, a tax is imposed by government, while an industry administered surcharge is more properly called a levy) be placed on broadband connections and recordable CDs, DVDs and BDs. As with many such taxes, or levies, this might be seen as unfair to those who do not indulge in the taxed behaviour.

What's wrong with DRM?

If DRM was always completely transparent to the end user, was not intrusive, protected copyright in a fashion that all agreed was fair and did not add cost to media hardware, perhaps no-one would find it objectionable. Various media companies have already applied almost draconian measures

A further problem with DRM is that of interoperability, that agreeing on and using a standard form of DRM is not always in the media companies interest. For example, this is true of Apple iTunes because of Apple’s agreements

with the content providers that iTunes downloads should not be easy to copy.

Some music download sites use license validation servers which validate a users right to play tracks they have downloaded. There have been several recent examples of music download sites closing down with the result that many users are left with tracks they have paid for but can no longer play because the validation servers are no longer available.

DRM adds complexity and expense to products that implement it. Although adding encryption to individual protected copies costs little, after all a necessary requirement from the replicators and copyright holders, it does affect both hardware, in the form of media players and software in the form of operating systems. A considerable amount of Microsoft’s development effort for the Vista OS went in to implementing DRM to the big media companies’ satisfaction. Vista incorporates what Microsoft call a Protected Environment (PE) to protect High Definition content. This creates a corridor through the operating system called a Protected Media Path (PMP), which, in theory, cannot be broken into. Device bus communications (which are accessible to users) associated with this process, are encrypted.

DRM has a negative effect on open source development, particularly on projects that are dedicated to pure open source, because all DRM involves proprietary methods and code. At present, playing protected Blu-ray content on a Linux box requires use of a device or player key, none of which have been issued for open source software players. However the keys for the X Box 360 and all of the popular Windows software players are now public knowledge and can be used by Linux players.

Open source opponents of DRM are often the most rabid. The Free Software Foundation (FSF) runs ‘Defective by Design’ campaigns and recently launched such an attack on Apple because the new MacBooks incorporate Apple’s DisplayPort connector, which includes the use of the optional DisplayPort Content Protection. Mac owners connecting an older Apple Cinema Display to view an iTunes movie will see an ‘un-authorised display’ message.

Perhaps the biggest disadvantage of DRM to the media companies that implement it is the alienation of customers and the bad publicity it has so far caused.

DRM in action

The exact form of DRM used depends on the media format and even on the distribution method. Music distribution is moving towards compressed audio file download over the internet, although audio CDs still form a large part of sales. Movies are still mainly sold on disk, either DVD or Blu-ray, and broadcast video is

distributed via terrestrial transmission, satellite or cable. Video is appearing on the internet, but for high quality video download or streaming to become a viable reality, a huge increase in bandwidth and capacity will be required. ISPs are already complaining that streaming services, such as the BBCs iPlayer, are causing network congestion.

It has been said that, despite trumpeting the high quality of CD Audio when it was first introduced, some elements of the music business always felt nervous about providing the public with studio quality recordings that could be copied digitally with absolutely no loss in quality.

Music downloads – Apple iTunes

Apples iTunes is one of the most well known and arguably most successful, of the digital music download sites. It exists mainly to act as the service site for the various versions of Apple’s portable music player devices, such as the iPod and iPhone. iTunes client software is bundled with every iPod sold. Apple iTunes uses the Advanced Audio Coding (AAC) format for its downloadable audio files. AAC is a lossy compression format intended as the successor to MP3, that has been standardized by ISO and IEC, as part of the MPEG-2 & MPEG-4 specifications. The MPEG-2 standard contains several audio coding methods, including the MP3 coding scheme. Apple iTunes is an application designed to run on an internet-connected PC running either MS Windows or an Apple Mac OS (note there is no version for Linux, although it can be hacked). Apple uses a version of AAC that includes its proprietary DRM extensions it calls FairPlay. This puts iTunes subscribers in the peculiar position of downloading AAC files, with FairPlay DRM extensions, which will only play on their registered version of iTunes on their computer, on an iPod or on an iTunes capable phone. However it is possible to burn the AAC files to a CD-R as MP3 files. The

